Homework 6

1. Problems and applications

2.

a) A binding price floor is set above the equilibrium price. The price of cheese rises, quantity demanded falls and quantity supplied rises, so there is a surplus of cheese.

b) Yes. Producers sell less at a higher price. If demand for cheese is elastic, the fall in quantity outweighs the higher price, so total revenue falls.

c) Producers benefit, because they now sell all their cheese at the floor price. Taxpayers lose, because they pay for the surplus. Consumers are unaffected compared with the basic price floor, since they still pay the same price and buy the same quantity.

3.

a) Equilibrium price $8, quantity 6 million Frisbees.

b) The price floor is $10. The new market price is $10 and 2 million Frisbees are sold (only as many as buyers demand).

c) The price ceiling is $9, which is above the equilibrium price, so it is not binding. The price stays at $8 and 6 million Frisbees are sold.

5.

No. The burden of a tax does not depend on who legally pays it, but on the elasticities of labor supply and demand. A higher tax on firms lowers the wage firms are willing to pay, so part of it is passed on to workers. Since total payroll taxes increase, workers will most likely end up worse off, especially because labor supply is fairly inelastic and workers bear most of the burden.

10.

a) $2P = 300 - P$, so $P = \$100$ and $Q = 200$.

b) A ceiling of $\$90$ is below equilibrium, so it is binding. Price $\$90$, quantity supplied 180, quantity demanded 210: a shortage of 30.

c) A floor of $\$90$ is below equilibrium, so it is not binding. Price $\$100$, quantity 200, neither a shortage nor a surplus.

d) $2(P - 30) = 300 - P$, so $P = \$120$ and $Q = 180$. There is neither a shortage nor a surplus. Buyers pay $\$120$, sellers receive $\$90$, and the quantity falls from 200 to 180.

2. Short essay: Should the minimum wage be $15 an hour?

In the article, David Neumark argues that a $15 minimum wage is not the best way to help low-wage workers. Low wages are mainly caused by technological change and globalization, which have reduced the value of low-skilled work, and a minimum wage cannot change these forces. From the perspective of supply and demand, a minimum wage is a price floor in the labor market. If it is set above the equilibrium wage, the quantity of labor supplied exceeds the quantity demanded, which causes unemployment. Workers who keep their jobs earn more, but the least skilled may lose their jobs or be replaced by more skilled workers. Neumark also points out that the minimum wage is poorly targeted: many minimum-wage earners are teenagers from middle-income families, while most poor families have no workers at all. As a result, only a small part of the higher wages actually reaches poor families.

1. As an economist, would I prefer a minimum wage or an earned income tax credit?

I would prefer the earned income tax credit (EITC). It targets low-income

families directly, encourages more people to work instead of reducing employment, and is paid for through the tax system, mostly by people with the highest incomes. A minimum wage, in contrast, is paid by businesses in low-wage industries and by their customers, who are often poor themselves, and it risks destroying jobs for the least skilled.

2. As a politician, which platform would be easier to campaign on?

A higher minimum wage would be easier to campaign on. It is simple to explain, it seems to cost the government nothing, and voters see an immediate pay raise. Its costs, such as lost jobs and higher prices, are hidden and spread out. A more generous EITC requires higher taxes, which are unpopular, and it is harder for voters to understand.

In conclusion, a $15 minimum wage may help some workers, but it is a blunt instrument. Better-targeted policies such as the EITC can help low-income families more effectively while doing less harm to the least-skilled workers.